

CBSE Class 10 Maths -- All Formulas

CBSE | Class 10 | Mathematics | 2026

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Algebra

Name	Formula / Value	Note
Quadratic Formula	$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$	For $ax^2+bx+c=0$
Sum of roots	$\alpha + \beta = -b/a$	Vieta's formulas
Product of roots	$\alpha * \beta = c/a$	
Discriminant	$D = b^2 - 4ac$	$D>0$: two real roots
AP nth term	$a_n = a + (n-1)d$	
AP sum	$S_n = n/2 * [2a + (n-1)d]$	Also $S_n = n/2(a+l)$

Trigonometry

Name	Formula / Value
Pythagorean identity	$\sin^2(\theta) + \cos^2(\theta) = 1$
Identity 2	$1 + \tan^2(\theta) = \sec^2(\theta)$
Identity 3	$1 + \cot^2(\theta) = \operatorname{cosec}^2(\theta)$
sin 30 deg	1/2
cos 30 deg	$\sqrt{3}/2$
tan 45 deg	1
sin 60 deg	$\sqrt{3}/2$

Name	Formula / Value
cos 60 deg	1/2

Coordinate Geometry

Name	Formula / Value	Note
Distance formula	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$	
Section formula	$((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n))$	Internal division
Mid-point	$((x_1 + x_2)/2, (y_1 + y_2)/2)$	
Area of triangle	$(1/2) x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) $	
Slope	$m = (y_2 - y_1)/(x_2 - x_1)$	

Mensuration

Name	Formula / Value	Note
Area of circle	$A = \pi r^2$	
Circumference	$C = 2\pi r$	
Volume of cylinder	$V = \pi r^2 h$	
Volume of cone	$V = (1/3)\pi r^2 h$	
Volume of sphere	$V = (4/3)\pi r^3$	
CSA of cylinder	$CSA = 2\pi r h$	
TSA of cylinder	$TSA = 2\pi r (r + h)$	
CSA of cone	$CSA = \pi r l$	$l = \text{slant height}$

Statistics & Probability

Name	Formula / Value	Note
Mean (grouped)	$\bar{x} = \frac{\sum(f_i \cdot x_i)}{\sum(f_i)}$	
Median (grouped)	$l + \left(\frac{n/2 - cf}{f}\right) \cdot h$	
Mode (grouped)	$l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \cdot h$	
Empirical relation	Mode = 3 Median - 2 Mean	Approximate
Probability	$P(E) = \frac{\text{Favourable}}{\text{Total outcomes}}$	$0 \leq P \leq 1$
Complementary	$P(E') = 1 - P(E)$	

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